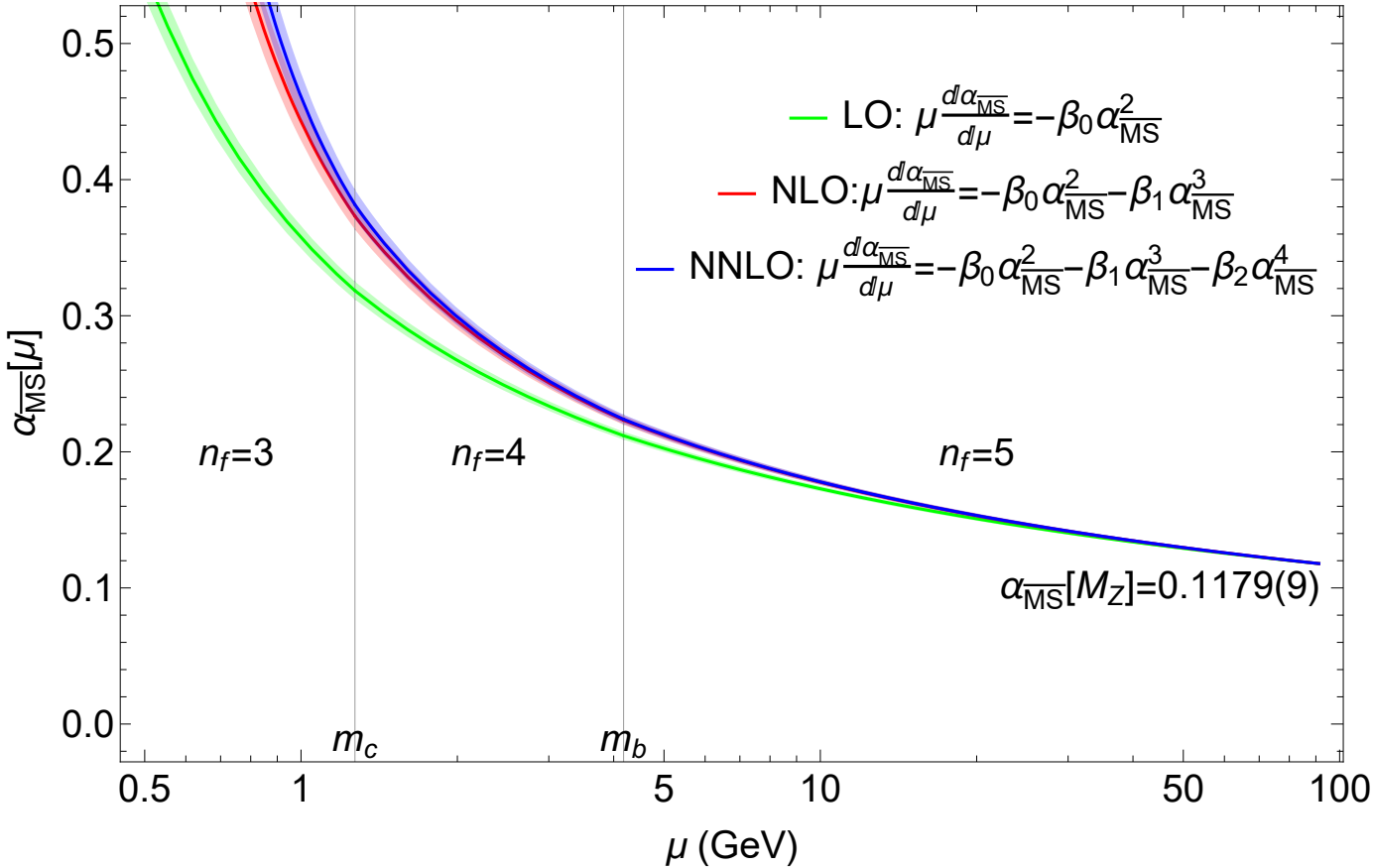
LO: $\mu \frac{d\alpha_{\overline{\rm MS}}}{d\mu} = -\beta_0 \alpha^2_{\overline{\rm MS}}$
NLO: $\mu \frac{d\alpha_{\overline{\rm MS}}}{d\mu} = -\beta_0 \alpha^2_{\overline{\rm MS}} - \beta_1 \alpha^3_{\overline{\rm MS}}$
NNLO: $\mu \frac{d\alpha_{\overline{\rm MS}}}{d\mu} = -\beta_0 \alpha^2_{\overline{\rm MS}} - \beta_1 \alpha^3_{\overline{\rm MS}} - \beta_2 \alpha^4_{\overline{\rm MS}}$
n_f=3
n_f=4
n_f=5
m_c
m_b
$\alpha_{\overline{\rm MS}}[M_Z]$=0.1179(9)
$\alpha_{\overline{\rm MS}}[\mu]$
μ (GeV)
0.0
0.1
0.2
0.3
0.4
0.5
0.5
1
5
10
50
100